# Network Security Investigation Report

**Date:** 8 January 2026 **Environment:** OPNsense Transparent Bridge IPS + UniFi Router

## Executive Summary

This document details a comprehensive security audit and IoT forensics investigation conducted on a home network. The investigation revealed Samsung Smart TV telemetry (ACR tracking) phoning home every 5 minutes, IoT devices bypassing local DNS, and successful implementation of mitigations including DNS-based blocking and NAT redirect rules.

# Part 1: Intrusion Prevention System (IPS) Audit

## 1.1 Network Topology

The network uses a transparent bridge configuration with OPNsense providing IPS capabilities:

*Internet → ISP Router (Modem Mode) → OPNsense Bridge (IPS) → UniFi Router → VLANs*

### OPNsense Configuration

| Component | Details |
| --- | --- |
| WAN Interface | igc0 [redacted] |
| LAN Management | 192.168[redacted] |
| DNS Interface | igc3 - 192.168.[redacted] (Unbound) |
| Mode | Transparent Bridge with Suricata IPS |
| IoT VLAN | 192.168.[redacted](managed by UniFi) |

## 1.2 Suricata IPS Alert Analysis (Past 7 Days)

Analysis of /var/log/suricata/eve.json* revealed 533 total alerts, with 7 blocked and 526 allowed.

### Alert Summary

| Alert Type | Count | Action | Assessment |
| --- | --- | --- | --- |
| ET POLICY ZIPPED EXE in transit | 515 | Allowed | Legitimate CDN downloads |
| CVE-2020-11900 Ripple20 exploit | 7 | BLOCKED | Inbound attacks blocked |
| TLSv1.0 deprecated protocol | 6 | Allowed | Legacy device |
| Reserved internal IP traffic | 3 | Allowed | Normal broadcast |

### Key Finding: Ripple20 Exploit Attempts

Seven CVE-2020-11900 (Ripple20) exploit attempts were successfully blocked. These attacks originated from known scanner hosting networks (Psychz Networks/FDCservers) targeting the WAN IP. This demonstrates the IPS is actively protecting the network from real-world threats.

**Source IPs blocked:** 64.62.156.146, 65.49.1.68, 65.49.1.164

### 1.3 IPS Performance Metrics

| Metric | Value | Status |
|---|---|---|
| Uptime | 6 days, 13 hours | Healthy |
| CPU Usage (Suricata) | 1.6% | Excellent |
| RAM Usage | 2.9GB / ~16GB (18.6%) | Plenty headroom |
| Packets Processed | 2.8 billion | - |
| Data Processed | 2.14 TB | - |
| Packet Drops | 178,990 (0.006%) | Static - OK |

The IPS is performing excellently with minimal resource usage and no performance issues detected.

# Part 2: Understanding IoT Callbacks

## 2.1 What Are IoT Callbacks?

IoT callbacks (also called "phone home" or "telemetry") are network connections initiated by smart devices to communicate with manufacturer servers. While some callbacks are necessary for device functionality, many are used for surveillance, advertising, and data collection without explicit user consent.

### Types of IoT Callbacks

1. **Legitimate callbacks:** Software updates, authentication, content delivery (Netflix streaming), push notifications
2. **Telemetry/Analytics:** Usage statistics, error reporting, performance metrics - often excessive
3. **Advertising/Tracking:** ACR (Automatic Content Recognition), ad networks, cross-device tracking
4. **Third-party sharing:** Data sent to Facebook, Google, ad exchanges without user knowledge

## 2.2 How ACR (Automatic Content Recognition) Works

ACR is a particularly invasive form of IoT callback used by Smart TVs. It works by:

- Capturing small audio samples or pixel regions from your screen every few seconds
- Creating unique fingerprints (hashes) - not actual screenshots or recordings
- Matching fingerprints against a database of known content
- Sending identified content back to manufacturer servers every few minutes
- Working across ALL inputs including HDMI (gaming consoles, streaming boxes)

**What ACR knows:** What you watch, when you watch, how long you watch, which HDMI input you use, channel changes, and pause/play behaviour.

**What ACR does NOT see:** Your network traffic, other devices, files, or browsing history. It is TV-specific surveillance.

# Part 3: IoT Forensics Investigation

## 3.1 Investigation Methodology

The investigation used DNS query logs from Unbound (running on OPNsense at 192.168.[redacted]) to identify IoT device behaviour. DNS logs reveal which domains devices are attempting to contact, providing visibility into callback activity.

**Log location:** /var/log/resolver/resolver_*.log

**Analysis period:** 1-8 January 2026

## 3.2 Devices Identified on IoT VLAN (192.168.[redacted])

| IP Address | Device | DNS Queries | Concern Level |
|---|---|---|---|
| 192.168.[redacted] | Apple TV / HomePod | 3,752 | Low - Normal |
| 192.168.[redacted] | Amazon Fire TV | 3,431 | Medium - DNS bypass |
| 192.168.[redacted] | Samsung Smart TV | 1,068+ | HIGH - ACR tracking |
| 192.168.[redacted] | Formuler Z11 Pro Max | 134 | Low - Android TV |
| 192.168.[redacted] | Netflix Device | 3,085 | Medium - Heavy logging |

## 3.3 Samsung TV - Detailed Analysis

The Samsung Smart TV was identified as the most concerning device, exhibiting heavy telemetry and tracking behaviour.

### Domains Contacted by Samsung TV

| Domain | Queries | Purpose |
|---|---|---|
| log-ingestion-eu.samsungacr.com | 286 | ACR viewing tracking |
| osb-v1-alb.samsungqbe.com | 96 | Telemetry |
| acr-eu-prd.samsungcloud.tv | 40 | Content recognition |
| tvx.adgrx.com | 50 | Ad tracking network |
| www.facebook.com | 32 | Facebook tracking |
| lcprd1.samsungcloudsolution.net | Various | Samsung cloud services |

### Key Finding: 5-Minute Phone Home Interval

Analysis of the DNS logs revealed the Samsung TV was contacting log-ingestion-eu.samsungacr.com approximately every 5 minutes, 24 hours a day. This represents Samsung's ACR system actively monitoring and reporting viewing habits regardless of whether the TV was being actively watched.

## 3.4 DNS Bypass Attempts

During investigation, it was discovered that some IoT devices were attempting to bypass the local DNS server (Unbound) by using hardcoded DNS servers, primarily Google (8.8.8.8).

**Affected device:** Amazon Fire TV (192.168.[redacted])

**Bypass method:** Hardcoded Google DNS (8.8.8.8) regardless of DHCP-assigned DNS

**Impact:** Device queries bypass Unbound, making blocking/logging ineffective

# Part 4: Remediation Actions

## 4.1 IoT VLAN DNS Configuration

The IoT VLAN DHCP was configured to assign the Unbound DNS server (192.168.[redacted]) to all IoT devices. This ensures DNS queries are logged and can be filtered.

**Location:** UniFi Controller → Networks → IoT VLAN → DHCP → DNS Server

**Value:** 192.168.[redacted]

## 4.2 DNS Redirect NAT Rule

To catch devices with hardcoded DNS (like Amazon Fire TV), a Destination NAT rule was created on UniFi to intercept and redirect all DNS traffic from the IoT VLAN to Unbound.

### NAT Rule Configuration

| Setting | Value |
|---|---|
| Type | Destination NAT |
| Interface | IoT VLAN (br10) |
| Protocol | TCP/UDP |
| Source Port | Any (CRITICAL - not 53) |
| Destination Port | 53 |
| Translated IP | 192.168.[redacted] |

**Verification:** iptables showed 2,077+ packets (141KB) successfully redirected

## 4.3 Samsung Tracking Domain Blocks

Five Samsung tracking domains were blocked using Unbound DNS Host Overrides. Each domain is returned as 0.0.0.0, preventing the TV from connecting to tracking servers.

### Blocked Domains

| Domain | Purpose Blocked |
|---|---|
| samsungacr.com | ACR viewing surveillance |
| samsungqbe.com | Samsung telemetry |
| samsungcloud.tv | Content recognition |
| samsungcloudsolution.net | Samsung cloud tracking |

| | |
|---|---|
| adgrx.com | Third-party ad tracking |

**Configuration location:** OPNsense → Services → Unbound DNS → Overrides

**Host value:** * (wildcard to catch all subdomains)

**IP value:** 0.0.0.0

### 4.4 Configuration Fix: QNAME Minimisation

During testing, YouTube and Google failed to resolve with SERVFAIL errors. This was caused by Strict QNAME Minimisation, a privacy feature that can break resolution for some domains when nameservers don't handle minimised queries properly.

**Fix:** Disable Strict QNAME Minimisation in OPNsense → Services → Unbound DNS → Advanced

**Note:** DNSSEC should remain ENABLED - only disable Strict QNAME Minimisation

# Part 5: Key Learnings

## 5.1 DNS as a Security Tool

DNS logging and filtering is one of the most effective tools for IoT security. By forcing all devices to use a local DNS resolver:

- All DNS queries are logged, providing visibility into device behaviour
- Malicious or tracking domains can be blocked network-wide
- Devices cannot simply bypass blocks by using alternative DNS
- DNSSEC validation protects against DNS spoofing attacks

## 5.2 IoT Devices Cannot Be Trusted

This investigation demonstrated that IoT devices routinely:

- Attempt to bypass user-configured DNS settings
- Send telemetry data without clear user consent
- Share data with third parties (Facebook, ad networks)
- Operate tracking systems 24/7 regardless of user activity

**Recommendation:** Always isolate IoT devices on a separate VLAN with restricted access to other network segments.

## 5.3 Defence in Depth

Effective IoT security requires multiple layers:

5. **VLAN Segmentation:** Isolate IoT from trusted devices
6. **DNS Control:** Force all DNS through a controlled resolver
7. **DNS Blocking:** Block known tracking domains
8. **NAT Redirect:** Catch devices that try to bypass DNS
9. **IPS/IDS:** Monitor and block malicious traffic patterns
10. **Logging:** Maintain visibility for forensics and auditing

## 5.4 Summary of Actions Taken

| Action | Status | Verified |
|---|---|---|
| Suricata IPS audit completed | Complete | Yes |
| IoT VLAN DNS set to Unbound | Complete | Yes |
| DNS redirect NAT rule created | Complete | 2077+ packets |

| | | |
|---|---|---|
| Samsung tracking domains blocked (5) | Complete | All 0.0.0.0 |
| QNAME Minimisation issue resolved | Complete | DNS working |

# Part 6: Future Recommendations

## Short Term

- Block DoT (DNS over TLS) on port 853 to prevent encrypted DNS bypass
- Block known DoH (DNS over HTTPS) provider domains
- Implement IoT VLAN isolation rules to prevent IoT accessing other VLANs

## Medium Term

- Add ET Pro Telemetry ruleset to Suricata (free upgrade from ET Open)
- Consider enabling DNSBL (DNS blocklist) in Unbound for broader ad/tracker blocking
- Disable Samsung ACR in TV settings (Settings → Support → Terms & Policies → Viewing Information Services)

## Long Term

- Set up automated alerting for suspicious DNS patterns
- Implement GeoIP blocking for traffic from high-risk countries
- Regular security audits of IPS alerts and DNS logs
- Document all firewall and DNS rules for future reference

*--- End of Report ---*